



MASTER 48-DATASET DIRECTORY FOR LAUNCH FEATURES & VOICE PIPELINE

This directory provides **48 curated open-source datasets** (6 datasets across each of HyrLink's 8 baseline launch features). It expands training coverage for resume transformation, deep probing, skill scoring, role-fit matchmaking, candidate proxy chat, and natural voice interaction (TTS prosody alignment & STT disfluency cleanup).

I. Dataset Download Directory (48 Sources)

Feature 1: Resume Rewriting (Narrative Format)

- **callmeech / resume-score-details**
<https://huggingface.co/datasets/callmeech/resume-score-details>
- **WikiBio Dataset (Infobox to Narrative Bio)**
https://huggingface.co/datasets/wiki_bio
- **ResumeAtlas (13.3k Classified Experience Arcs)**
<https://huggingface.co/papers/2402.03832>
- **skepss / resume-job-description-fit**
<https://huggingface.co/datasets/skepss/resume-job-description-fit>
- **DatasetMaster Normalized Resumes (JSON Schema)**
<https://huggingface.co/datasets/datasetmaster/resumes>
- **Saugataroyarghya / Resume Dataset**
<https://www.kaggle.com/datasets/saugataroyarghya/resume-dataset>

Feature 2: Candidate Deep Dives (Questionnaire & Probing)

- **AnthropicInterviewer (1,250 Qualitative AI Transcripts)**
<https://huggingface.co/datasets/Anthropic/AnthropicInterviewer>
- **Inquisitive Question Generation Dataset**
<https://huggingface.co/datasets/inquisitive>
- **CounselChat (Active Listening & Reflective Follow-ups)**
<https://huggingface.co/datasets/nbertagnolli/counsel-chat>
- **Comprehensive QA Benchmark (CoQA & QuAC Multi-turn)**
<https://huggingface.co/datasets/Successmove/comprehensive-qa-dataset>
- **QASC (Question Answering via Sentence Composition)**
<https://huggingface.co/datasets/allenai/qasc>
- **MentalChat16K (Behavioral & Clinical Probing)**
<https://huggingface.co/datasets/ShenLab/MentalChat16K>

Feature 3: Skill Validation & Testing

- **SkillSpan (Hard & Soft Skill Extraction)**
<https://huggingface.co/datasets/jjzha/skillspan>
- **TechWolf Skill Extraction (ESCO Span Benchmark)**
<https://huggingface.co/datasets/TechWolf/skill-extraction-tech>
- **MBPP (Mostly Basic Python Problems)**

<https://huggingface.co/datasets/google-research-datasets/mbpp>

■ **OpenAI HumanEval (Code Skill Execution)**

https://huggingface.co/datasets/openai/openai_humaneval

■ **SWE-bench (Software Engineering Problem Solving)**

<https://huggingface.co/datasets/princeton-nlp/SWE-bench>

■ **APPS Dataset (10,000 Coding Challenges)**

<https://huggingface.co/datasets/codeparrot/apps>

Feature 4: Role-Fit Scoring

■ **Candidate Job Role & Skill Gap Dataset**

<https://www.kaggle.com/datasets/ckshetty/candidate-job-role-dataset>

■ **AI Recruitment Pipeline Dataset**

<https://www.kaggle.com/datasets/yaswanthkumary/ai-recruitment-pipeline-dataset>

■ **facehuggerapoorv / resume-jd-match**

<https://huggingface.co/datasets/facehuggerapoorv/resume-jd-match>

■ **netsol / resume-score-details**

<https://huggingface.co/datasets/netsol/resume-score-details>

■ **syncoraai / resume-screening-llm-training-dataset**

<https://www.kaggle.com/datasets/syncoraai/resume-screening-llm-training-dataset>

■ **sonalshinde123 / ai-driven-resume-screening-dataset**

<https://www.kaggle.com/datasets/sonalshinde123/ai-driven-resume-screening-dataset>

Feature 5: Pro Chatbot (AI Proxy of Candidate)

■ **Persona-Chat (Trait-Grounded Dialogue)**

https://huggingface.co/datasets/facebook/persona_chat

■ **EmpatheticDialogues (Emotional Context Dialogue)**

https://huggingface.co/datasets/facebook/empathetic_dialogues

■ **DailyDialog (Multi-turn Natural Conversation)**

https://huggingface.co/datasets/daily_dialog

■ **Blended Skill Talk (Multi-skill Persona Conversations)**

https://huggingface.co/datasets/facebook/blended_skill_talk

■ **OpenAssistant / oasst1 (Human-annotated Instruction Follow)**

<https://huggingface.co/datasets/OpenAssistant/oasst1>

■ **Synthetic Persona Chat (Multi-perspective Dialogue)**

<https://huggingface.co/datasets/google/synthetic-persona-chat>

Feature 6: Recruiter Job Postings & Candidate Pairing

■ **Candidate Recommendation Model Datasets**

<https://github.com/Srikanth-R/Candidate-Recommendation-Model>

■ **Salient Entity Extraction Dataset**

<https://github.com/acp19tag/skill-extraction-dataset>

■ **Skill Scoring NLP Model Repository**

<https://github.com/OfirOhan/skill-scoring-nlp-model>

■ **Resume Skills vs Interview Calls Dataset**

<https://www.kaggle.com/datasets/prince7489/resume-skills-vs-interview-calls-dataset>

■ **Skill-Evol-Bench (Skill Evolution & Benchmark)**

<https://huggingface.co/datasets/SkillEvolBench-Team/Skill-Evol-Bench>

■ **amjad-awad / skill-extractor**

<https://huggingface.co/amjad-awad/skill-extractor>

Feature 7: Voice Synthesization (TTS Alignment & Prosody)

- **DisfluencySpeech (Conversational Prosody & Paralanguage)**
<https://huggingface.co/datasets/amaai-lab/DisfluencySpeech>
- **SeamlessExpressive / Espresso (Expressive Prosody & Pitch)**
<https://huggingface.co/facebook/seamless-expressive>
- **LibriTTS-R (Restored Clean Audio & Duration Alignments)**
https://huggingface.co/datasets/parler-tts/libritts_r_filtered
- **LJSpeech (Single Speaker Benchmark Corpus)**
https://huggingface.co/datasets/lj_speech
- **CSTR VCTK Corpus (Multi-accent Speech Corpus)**
<https://huggingface.co/datasets/cstr-edinburgh/vctk>
- **MELD (Multimodal Emotion Lines Dataset)**
<https://huggingface.co/datasets/MELD>

Feature 8: Speech-to-Text (STT) Processing & Cleanup

- **ami-disfluency (Annotated Meeting Spoken Transcripts)**
<https://huggingface.co/datasets/hhoangphuoc/ami-disfluency>
- **Switchboard Disfluency Transcripts (switchboard_dps)**
https://huggingface.co/datasets/switchboard_dps
- **Transcription Cleanup Trainer (Raw vs Clean Pairs)**
<https://huggingface.co/datasets/danielrosehill/Transcription-Cleanup-Trainer>
- **Auto Whisper Disfluency Dataset**
<https://huggingface.co/datasets/pradachan/auto-whisper-disfluency>
- **VoxPopuli (Spontaneous Speech Transcripts)**
<https://huggingface.co/datasets/facebook/voxpopuli>
- **LibriSpeech ASR (Clean & Other Transcripts)**
https://huggingface.co/datasets/librispeech_asr

II. Dataset Summaries & Use Cases

Feature 1: Resume Rewriting (Narrative Format)

1. callmeech / resume-score-details

CATEGORY: NARRATIVE RESUME TRANSFORMATION

Summary: 1,031 paired resumes and JDs generated with GPT-4o containing rich micro/macro justification narratives.
Use Case: Teaches the SLM to convert dry experience bullets into persuasive narrative summaries.

2. WikiBio Dataset

CATEGORY: TABULAR TO PROSE SYNTHESIS

Summary: 728,321 Wikipedia infobox tables mapped directly to introductory narrative biographical paragraphs.
Use Case: Fine-tunes structured-data-to-narrative generation, preserving exact facts without dropping metrics.

3. ResumeAtlas

CATEGORY: SENIORITY NARRATIVE ALIGNMENT

Summary: 13,389 resumes with multi-round human expert annotations mapped to experience tiers.
Use Case: Rewrites candidate bullet points into executive narrative stories matching target seniority.

4. skepss / resume-job-description-fit

CATEGORY: RESUME CONTEXTUAL FRAMING

Summary: Contextually paired resumes and job descriptions emphasizing career trajectory alignment.

Use Case: Teaches the SLM to frame a candidate's historical experience around the target role's narrative arc.

5. DatasetMaster Normalized Resumes

CATEGORY: CLEAN JSON PARSING

Summary: Clean JSON schema normalization of raw candidate resumes.

Use Case: Provides clean input schema structuring before sending extracted data into narrative generation prompts.

6. Saugataroyarghya / Resume Dataset

CATEGORY: MULTI-DISCIPLINE SUMMARIZATION

Summary: 10,000 resumes across 25 career domains paired with candidate objectives.

Use Case: Expands domain-specific vocabulary for rewriting engineering, marketing, finance, and product resumes.

Feature 2: Candidate Deep Dives (Questionnaire & Probing)

7. AnthropicInterviewer

CATEGORY: QUALITATIVE AI PROBING

Summary: 1,250 qualitative, semi-structured interviews conducted by AI.

Use Case: Fine-tunes the SLM on how to ask deep follow-up questions based on unpolished candidate answers.

8. Inquisitive Question Generation

CATEGORY: ELICITATIVE QUESTIONING

Summary: 19,000 text spans paired with questions designed to evoke deeper explanations and underlying motivations.

Use Case: Generates targeted follow-up questions when a candidate provides brief or ambiguous answers.

9. CounselChat

CATEGORY: REFLECTIVE LISTENING

Summary: Counseling dialogues focusing on reflective active listening and underlying context extraction.

Use Case: Teaches HyrLink to uncover root candidate experiences and core achievements behind unpolished answers.

10. Comprehensive QA Benchmark

CATEGORY: MULTI-TURN QA LOGIC

Summary: 160k+ QA pairs combining CoQA and QuAC multi-turn conversational dialogs.

Use Case: Maintains contextual memory over multi-step questionnaire deep dives with candidates.

11. QASC (Sentence Composition)

CATEGORY: FACT SYNTHESIS

Summary: Multi-fact reasoning dataset combining separate facts into unified explanations.

Use Case: Synthesizes multiple scattered candidate answers into one coherent career story.

12. MentalChat16K

CATEGORY: BEHAVIORAL PROBING

Summary: 16,000 multi-turn behavioral dialogues focused on personal narrative extraction.

Use Case: Probes candidate soft skills, workplace challenge handling, and interpersonal style.

Feature 3: Skill Validation & Testing

13. SkillSpan

CATEGORY: SKILL EXTRACTION

Summary: ESCO-mapped hard/soft skill span annotations from job texts.

Use Case: Extracts granular skills from candidate claims to isolate what needs to be validated.

14. TechWolf Skill Extraction

CATEGORY: TECHNICAL SKILL BOUNDARIES

Summary: Sentence and span-level skill tags across technical job postings.

Use Case: Defines technical boundaries so HyrLink generates domain-accurate testing scenarios.

15. MBPP (Mostly Basic Python Problems)

CATEGORY: ENTRY/MID SKILL VALIDATION

Summary: 974 programming problems with task descriptions, code solutions, and test cases.

Use Case: Evaluates entry-level technical candidate responses with automated test case validation.

16. OpenAI HumanEval

CATEGORY: LOGIC VALIDATION

Summary: 164 hand-crafted coding problems with unit tests and docstrings.

Use Case: Generates dynamic live-coding scenario prompts and evaluates code correctness.

17. SWE-bench

CATEGORY: COMPLEX SYSTEM VALIDATION

Summary: Real software engineering GitHub issues, code patches, and test suite executions.

Use Case: Validates senior technical claims by setting up realistic engineering problem-solving scenarios.

18. APPS Dataset

CATEGORY: COMPREHENSIVE TECHNICAL TESTING

Summary: 10,000 coding problems ranging from introductory to competitive programming.

Use Case: Calibrates numeric skill validation scores based on problem difficulty tier.

Feature 4: Role-Fit Scoring

19. Candidate Job Role & Skill Gap Dataset

CATEGORY: GAP ANALYSIS

Summary: Candidate profiles paired with target job roles and explicit skill gap annotations.

Use Case: Directly trains the SLM to output "What's Missing" reports and weighted gap scores.

20. AI Recruitment Pipeline Dataset

CATEGORY: OUTCOME-CALIBRATED MATCHING

Summary: Resume and screening data linked to actual final hiring decisions.

Use Case: Calibrates role-fit scoring models against real recruitment selection outcomes.

21. facehuggerapoorv / resume-jd-match

CATEGORY: SEMANTIC FIT RATING

Summary: Paired resume and JD samples with semantic match scores.

Use Case: Fine-tunes overall profile alignment scoring logic.

22. netsol / resume-score-details**CATEGORY: MACRO/MICRO WEIGHTING****Summary:** Resumes scored against JDs with explicit macro (experience/strategy) and micro (skill) weighting.**Use Case:** Teaches HyrLink to generate structured weighted scoring rubrics and justification paragraphs.**23. syncoraai / resume-screening-llm-training****CATEGORY: SYNTHETIC SCREENING INSTRUCTION****Summary:** Instruction dataset for candidate evaluation and ranking.**Use Case:** Enhances instruction following for complex multi-criteria candidate screening prompts.**24. sonalshinde123 / ai-driven-resume-screening****CATEGORY: MULTI-CATEGORY SCREENING****Summary:** Multi-category candidate screening dataset with explicit qualification badges.**Use Case:** Improves hard skill and qualification check accuracy during scoring.**Feature 5: Pro Chatbot (AI Proxy of Candidate)****25. Persona-Chat****CATEGORY: TRAIT GROUNDING****Summary:** Multi-turn dialogues where agents adhere strictly to assigned background facts and tone.**Use Case:** Ensures the Pro Chatbot maintains candidate personality and career facts in recruiter interviews.**26. EmpatheticDialogues****CATEGORY: EMOTIONAL ALIGNMENT****Summary:** 25k conversations grounded in emotional contexts.**Use Case:** Infuses warmth, confidence, and natural emotional tone into candidate proxy responses.**27. DailyDialog****CATEGORY: NATURAL DIALOGUE FLOW****Summary:** High-quality multi-turn dialogues covering daily life and professional interactions.**Use Case:** Fine-tunes conversational turn-taking, cadence, and backchanneling.**28. Blended Skill Talk****CATEGORY: MULTI-SKILL PERSONA SWITCHING****Summary:** Dialogues combining persona adherence, knowledge sharing, and empathy.**Use Case:** Enables the proxy chatbot to discuss technical skills while staying in candidate persona.**29. OpenAssistant / oasst1****CATEGORY: HUMAN INSTRUCTION FOLLOWING****Summary:** Human-annotated conversational instruction dataset.**Use Case:** Improves responsiveness to unexpected recruiter questions during AI proxy interviews.**30. Synthetic Persona Chat****CATEGORY: EXTENDED PERSONA CONSISTENCY****Summary:** Extended multi-turn dialogues featuring detailed character profiles.**Use Case:** Prevents persona degradation during long recruiter chat sessions.

Feature 6: Recruiter Job Postings & Candidate Pairing

31. Candidate Recommendation Model Datasets

CATEGORY: POOL MATCHMAKING

Summary: Frameworks mapping recruiter job requisitions to candidate pools.

Use Case: Ranks candidate databases against new job postings and picks top choices.

32. Salient Entity Extraction Dataset

CATEGORY: REQUISITION PARSING

Summary: Labeled entity spans in job text (skills, domain, experience).

Use Case: Parses recruiter job postings into structured search queries.

33. Skill Scoring NLP Model Repository

CATEGORY: CANDIDATE RANKING LOGIC

Summary: Semantic evaluation metrics for scoring candidate skills against job requirements.

Use Case: Drives comparative candidate selection rationales.

34. Resume Skills vs Interview Calls

CATEGORY: CALL-BACK PROBABILITY CALIBRATION

Summary: Resume feature sets correlated with actual interview invitation rates.

Use Case: Calibrates candidate pairing logic toward profiles most likely to get interview calls.

35. Skill-Evol-Bench

CATEGORY: SKILL TAXONOMY MAPPING

Summary: Evolving skill benchmark mapping legacy skill names to modern tech stack equivalents.

Use Case: Matches candidate profiles with older skill phrasing to modern job descriptions.

36. amjad-awad / skill-extractor

CATEGORY: RAPID SKILL EXTRACTION

Summary: Lightweight skill extraction model and training corpus.

Use Case: Powers fast candidate filtering across large resume databases.

Feature 7: Voice Synthesization (TTS Alignment & Prosody)

37. DisfluencySpeech

CATEGORY: CONVERSATIONAL PARALANGUAGE

Summary: Studio conversational English speech dataset with annotated non-speech events and pause markers.

Use Case: Inserts natural paralinguage markers ('[laughter]', '[sigh]', '[pause]') into TTS text.

38. SeamlessExpressive / Espresso

CATEGORY: PITCH & PROSODY MAPPING

Summary: High-quality expressive speech annotated with pitch contours, speech rate, and emotion styles.

Use Case: Teaches the SLM to output SSML prosody tags to control downstream voice style.

39. LibriTTS-R

CATEGORY: RESTORED CLEAN PROSODY

Summary: 585 hours of clean restored audio with normalized text and phonetic alignments.

Use Case: Fine-tunes phonetic spelling hints and text normalization for crisp pronunciation.

40. LJSpeech

CATEGORY: SINGLE-SPEAKER ACOUSTIC BASELINE

Summary: 13,100 clean single-speaker audio clips with exact transcriptions.

Use Case: Provides acoustic baseline alignment for clean text-to-phoneme output generation.

41. CSTR VCTK Corpus

CATEGORY: MULTI-ACCENT PROSODY

Summary: Speech recordings from 110 English speakers across diverse regional accents.

Use Case: Prepares the SLM to format text for diverse candidate voice clones and accents.

42. MELD (Multimodal Emotion Lines)

CATEGORY: CONVERSATIONAL EMOTION TAGS

Summary: Conversational dialogue audio tagged with emotion labels (joy, surprise, fear, neutral).

Use Case: Maps candidate conversational intent to explicit emotional tags for expressive TTS rendering.

Feature 8: Speech-to-Text (STT) Processing & Cleanup

43. ami-disfluency

CATEGORY: SPOKEN DISFLUENCY ANNOTATIONS

Summary: 35,000+ annotated disfluency segments (false starts, stutters, filler words) in meeting audio.

Use Case: Fine-tunes the SLM to clean unedited spoken transcriptions into clear text.

44. Switchboard Disfluency Transcripts

CATEGORY: RAW VS CLEAN SPOKEN TEXT

Summary: Conversational transcripts with explicit disfluency edit tags.

Use Case: Strips filler words ("um/uh", false starts) while preserving candidate core intent.

45. Transcription Cleanup Trainer

CATEGORY: GOLDBLOCKS TRANSCRIPT CLEANUP

Summary: Paired raw Whisper transcripts and manually cleaned ground-truth targets.

Use Case: Directly trains the SLM to transform raw ASR audio transcripts into polished sentences.

46. Auto Whisper Disfluency Dataset

CATEGORY: ASR DISFLUENCY REDUCTION

Summary: Benchmark dataset paired for training ASR post-processing cleanup models.

Use Case: Cuts transcription artifacts and run-on sentences in spoken candidate answers.

47. VoxPopuli

CATEGORY: SPONTANEOUS SPEECH TRANSCRIPTS

Summary: Large-scale spontaneous speech transcripts with natural speech disfluencies.

Use Case: Improves SLM robustness against noisy, informal, or accented candidate audio transcriptions.

48. LibriSpeech ASR

CATEGORY: ACOUSTIC TEXT NORMALIZATION

Summary: 1,000 hours of read speech with normalized vs unnormalized text transcriptions.

Use Case: Handles number expansion, acronym normalization, and punctuation restoration in raw STT text.